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Assingment 2

1. Confidentiality

Confidentiality keeps sensitive data strictly in the hands of people cleared to see it, blocking unauthorized eyes.

1. Role-Based Access Control (RBAC): Setting up an Electronic Health Record system so nurses can check medical histories, while the billing team only sees invoice data and zero medical details.

2. Data Encryption at Rest: Locking down a database containing payment info with AES-256 encryption. If someone physically steals the server's hard drive, the files stay completely unreadable without the key.

3. Multi-Factor Authentication (MFA): Asking staff to enter a temporary code from an authenticator app alongside their standard password when logging into company tools, stopping hackers who buy leaked passwords.

4. Data Loss Prevention (DLP) Policies: Running software that scans outgoing corporate emails and automatically halts any message containing raw credit card details or identity numbers.

2. Integrity

Integrity guarantees that information stays authentic, exact, and untouched—whether it’s sitting on a server or moving across a network.

1. File Hash Verification: Publishing an SHA-256 checksum next to a software download so users can check that their installer hasn't been tampered with or swapped for malware.

2. Database Constraints & Transactions: Wrapping financial updates in strict database rules so if a bank transfer drops mid-way, the entire transaction rolls back instead of leaving funds missing.

3. Digital Signatures: Attaching a public-key digital signature to contract files, which immediately breaks and alerts everyone if someone edits the document text after signing.

4. Append-Only System Logs: Writing administrative events to write-once logs that prevent anyone—even an admin—from deleting or altering past records to cover their tracks.

3. Availability

Availability makes sure that systems, networks, and services stay up and running for users when they need them.

1. Load Balancing Across Servers: Splitting web traffic across several redundant servers so if one box crashes, the rest handle the load without dropping the site offline.

2. Offsite Backups & Recovery Runs: Keeping encrypted daily backups on a remote server and running regular restoration drills so the business can recover fast from a ransomware attack.

3. DDoS Protection Services: Routing network traffic through scrubbing tools (like Cloudflare or AWS Shield) to filter out fake request surges before they overwhelm the web servers.

4. Data Center Battery Banks & Generators: Running battery UPS units and backup diesel generators at server facilities to prevent sudden shutdowns during neighborhood power outages.